

### Exercise 1

Show that  $Cov[x, x]$  is symmetric and positive definite.

Since each  $X_i$  is random real number:

$$[Cov[x, x]]_{ij} = E[X_i X_j] - E[X_i]E[X_j] = E[X_j X_i] - E[X_j]E[X_i] = [Cov[x, x]]_{ji}$$

Therefore  $Cov[X, X]$  is symmetric

$$\begin{aligned} \sum_{ij} \alpha_i \alpha_j [Cov[X, X]]_{ij} &= \sum_{ij} \alpha_i \alpha_j (E[X_i X_j] - E[X_i]E[X_j]) = \\ &= \sum_{ij} \alpha_i \alpha_j E[X_i X_j] - \sum_i \alpha_i E[X_i] \sum_j \alpha_j E[X_j] = E \left[ \sum_i \alpha_i X_i \sum_j \alpha_j X_j \right] - E \left[ \sum_i \alpha_i X_i \right] E \left[ \sum_j \alpha_j X_j \right] = \end{aligned}$$

$E[Y^2] - (E[Y])^2 \geq 0$  by Jensen's inequality.

### Exercise 2

Let  $X \sim MVN(\mu, \Sigma)$  and  $\Sigma = U \Lambda U^T$  and let  $Y_i = (X - \mu)^T u_i$

where the  $u_i$  are columns of  $U$ .

1. Show that the  $Y_i$  as defined are independent.

Since  $X$  is normally distributed this implies that  $Y$  is as well. Hence it is sufficient to show that for each  $i$  and  $j$ ,  $Y_i$  and  $Y_j$  are uncorrelated.

$$[Cov[Y_i, Y_j]]_{ij} = Cov[Y_i, Y_j] = Cov[(X - \mu)^T u_i, (X - \mu)^T u_j] = u_i Cov[(X - \mu)^T, (X - \mu)^T] u_j = u_i \Sigma u_j = u_i \lambda_j u_j = \lambda_j \delta_{ij} \implies Y_i \text{ and } Y_j \text{ are uncorrelated and hence independent.}$$

2. Show that  $Y_i \sim N(0, \lambda_i)$

$$\text{Since } E[Y_i] = (E[X] - \mu)^T u_i = (\mu - \mu)^T u_i = 0 \text{ and } 1. \implies (diag(\Sigma))_i = \lambda_i \implies Y_i \sim N(0, \lambda_i)$$

3. Show  $X$  can be written in the form  $X = \mu + Y_1 u_1 + \dots + Y_k u_k$

$$E[X] = E[\mu] + E[Y_1] u_1 + \dots + E[Y_k] u_k = \mu$$

With  $U = (u_1, \dots, u_k)$  we have

$$Var[X] = Var[UY + \mu] = Cov[UY + \mu, UY + \mu] = Cov[UY, UY] = U Cov[Y, Y] U^T =$$

$$U \begin{bmatrix} \lambda_0 & & \\ & \ddots & \\ & & \lambda_k \end{bmatrix} U^T = \Sigma$$

Hence  $X \sim MVN(\mu, \Sigma)$

### Exercise 3

#### Bayesian Inference

1. Show that  $X \sim \text{Beta}(\alpha, \beta) \implies P(X|y) = \text{Beta}(a, b)$  for some a, b.

Let  $S_n = \sum_{i=1}^n$ . Then

$$\begin{aligned} P(x|y) &= \frac{P(y|x)P(x)}{P(y)} = \prod_{i=1}^n x^{y_i} (1-x)^{(1-y_i)} x^{(\alpha-1)} (1-x)^{(\beta-1)} \mathbb{1}_{[0,1]}(x) = \\ &= \left( x^{\sum y_i} (1-x)^{(n-\sum y_i)} \right) \left( x^{(\alpha-1)} (1-x)^{(\beta-1)} \right) \mathbb{1}_{[0,1]}(x) = \\ &= x^{(\sum y_i + \alpha - 1)} (1-x)^{(n - \sum y_i + \beta - 1)} \mathbb{1}_{[0,1]}(x) = x^{(a-1)} (1-x)^{(b-1)} \mathbb{1}_{[0,1]}(x) \text{ with } a = \sum y_i + \alpha \text{ and } \\ &= b = n - \sum y_i + \beta \end{aligned}$$

This implies

$$E[X|y] = \frac{a}{(a+b)} = \frac{(\sum y_i + \alpha)}{(\sum y_i + \alpha + n - \sum y_i + \beta)} = \frac{(n\bar{y} + \alpha)}{(\alpha + \beta + n)} \implies$$

$$w(n) = \frac{n}{(\alpha + \beta + n)} \text{ and } 1 - w(n) = \frac{(\alpha + \beta)}{(\alpha + \beta + n)} \text{ and}$$

$$E[X|y] = w\bar{y} + (1-w)\alpha/(\alpha + \beta)$$

#### Exercise 4

$$X \sim N(\mu_0, \lambda_0^{-1}), Y_i = X + \epsilon_i, \epsilon_i \sim N(0, \lambda^{-1}), D = \lambda_0/\lambda$$

1. Show that  $\left(I + \frac{\mathbb{I}\mathbb{I}^T}{D}\right)^{-1} = I - (n + D)^{-1} \mathbb{I}\mathbb{I}^T$

Using the Sherman-Morrison formula with  $u = \frac{\mathbb{I}}{D}$  and  $v = \mathbb{I}$  gives

$$\begin{aligned} (I + uv^T)^{-1} &= I - \frac{uv^T}{1 + v^T u} = I - \frac{uv^T}{(1 + n/D)} = I - \left( \frac{1}{D(1 + n/D)} \right) \mathbb{I}\mathbb{I}^T = \\ &= I - (n + D)^{-1} \mathbb{I}\mathbb{I}^T \end{aligned}$$

2. Verify that  $E[X|y] = w\mu_0 + (1-w)\bar{y}$  with  $w = \lambda_0(n\lambda_0 + \lambda)^{-1}$

$$\Sigma_{11} = \lambda_0^{-1}, \Sigma_{12} = \lambda_0^{-1} \mathbb{I}^T,$$

$$\Sigma_{22} = \begin{bmatrix} \lambda_0^{-1} + \lambda^{-1} & & & & \lambda_0^{-1} \\ & \ddots & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & \ddots \\ & & \lambda_0^{-1} & & & \lambda_0^{-1} + \lambda^{-1} \end{bmatrix}$$

And using the inversion formula above gives

$$\begin{aligned} \Sigma_{22}^{-1} &= (\lambda^{-1} I + \lambda_0^{-1} \mathbb{I}\mathbb{I}^T)^{-1} = (\lambda^{-1} (I + \frac{1}{D} \mathbb{I}\mathbb{I}^T))^{-1} = \\ &= \lambda (I - (n + D)^{-1} \mathbb{I}\mathbb{I}^T) \end{aligned}$$

$$\begin{aligned}\Sigma_{12}\Sigma_{22}^{-1}y &= \lambda_0^{-1}\mathbb{I}^T\lambda\left(I - (n+D)^{-1}\mathbb{I}\mathbb{I}^T\right)y = \\ \frac{\mathbb{I}^Ty}{D} - \frac{\mathbb{I}^T\mathbb{I}\sum y_i}{D(n+D)} &= \frac{\sum y_i}{D} - \frac{n\sum y_i}{D(n+D)} = \frac{n\bar{y}}{D} - \frac{n^2\bar{y}}{D(n+D)} = \\ \frac{n\bar{y}}{D}\left(1 - \frac{n}{n+D}\right) &= \frac{n\bar{y}}{D}\left(\frac{D}{n+D}\right) = \bar{y}\frac{n}{(n+D)} = \frac{n\bar{y}\lambda}{(n\lambda + \lambda_0)}\end{aligned}$$

$$\text{Setting } 1 - w = \frac{n\lambda}{(n\lambda + \lambda_0)} \implies w = \frac{D}{(n+D)} = \frac{\lambda_0}{(n\lambda + \lambda_0)} \implies$$

$$\hat{X} = w(n)\mu_0 + (1 - w(n))\bar{y}$$

$$Cov[X|y] = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}^T$$

$$\begin{aligned}\Sigma_{11} &= \lambda_0^{-1}, \Sigma_{12} = \lambda_0^{-1}\mathbb{I}^T, \\ \Sigma_{22}^{-1} &= \lambda\left(I - (n+D)^{-1}\mathbb{I}\mathbb{I}^T\right) \implies\end{aligned}$$

$$\begin{aligned}Cov(X|y) &= \lambda_0^{-1} - \lambda_0^{-1}\mathbb{I}^T\lambda\left(I + \frac{1}{D}\mathbb{I}\mathbb{I}^T\right)^{-1}\lambda_0^{-1}\mathbb{I} = \\ \lambda_0^{-1} - \frac{\lambda_0^{-1}\mathbb{I}^T\mathbb{I}}{D} + \frac{\lambda_0^{-1}\mathbb{I}^T\mathbb{I}\mathbb{I}^T\mathbb{I}}{D(n+D)} &= \lambda_0^{-1}\left(1 - \frac{n}{D} + \frac{n^2}{D(n+D)}\right) = \\ \lambda_0^{-1}\left(\frac{nD + D^2 - n^2 - nD + n^2}{D(n+D)}\right) &= \lambda_0^{-1}\left(\frac{D}{(n+D)}\right) = \lambda_0^{-1}\left(\frac{\lambda_0}{n\lambda + \lambda_0}\right) = \lambda_0^{-1}w\end{aligned}$$

$$Var[X|y] = Cov[X|y, X|y] = Cov[\lambda_0^{-1}w, \lambda_0^{-1}w] = \lambda_0^{-1}w$$